
PrimeCore Minerals

P5.4 Test network performance under heavy traffic

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Introduction

A reliable network must be able to withstand being under load when ports are maxed out at their top speed and not only being able to stay on but also process all the packets accordingly. This was determined early in the project and is number P5.4 on our scope.

The three main tests to confirm the reliability of a network include latency, bandwidth speed and jitter.

Latency confirms how long it takes network packets to travel between the 2 network devices with the lower the number the better. It is measured in milliseconds.

Bandwidth demonstrates the maximum speed a port can maintain.

Jitter shows the time variation between packets arriving. It is also known as network shaking and while some is expected, a large jitter shows congestion and packets arriving out of order, which will take more time for the switch to organise and therefore putting the switch under load.

A reliable network will allow the security/penetration tests to be conducted successfully as the network reliability will not affect the results.

Purpose

To test this scenario, a port limit of 100Mbps for standard ports and 1Gbps for trunk ports was placed on the LAN switches. This limit is lower than the switches recommended actual speed but is placed at this to protect the virtual environment from crashing.

Ping is the command that will be used to show latency between switches and is built into linux.

To demonstrate the analogy of the ports under maximum load, iperf3 has been installed on the switches and APs on the site side to simulate the load.

Tests have been completed from each switch to demonstrate that each switch is able to withstand the ability of being under load.

Iperf3 command and switches used in testing

The commands used to run the tests are as follows.

1. Receiving switch
 - a. iperf3 -s

2. Sending switch

- a. To run maximum speed test with TCP packets
 - i. Iperf3 -t 40 -c <receiving switch IP address>
- b. To test for jitter using UDP packets
 - i. Iperf3 -u -b 0 -l 1400 -t 40 -c <receiving switch IP address>

The iperf3 online documentation can be found here <https://iperf.fr/iperf-doc.php> with a list of the switches used below

-s = server (in this project switch/AP) receiving the results and replying

-t = time taken to run the test shown in seconds

-c = server (in this project switch/AP) sending the results and controlling the test

-u = use UDP packets only. This is to flood the network

-b 0 = bandwidth of 0 means flood the network as much as you can

-l 1400 = limit packet size to 1400 bytes. Network standard is 1500 bytes so this is below that to prevent packet fragmentation

Switch Speed Test Results

The following table is an overview of the results from the tests

Switch/AP Name	Ping/Speed/Jitter Test	Results
PCM-SITEA-SW01	Ping Test	0.680ms 0 packet loss
PCM-SITEA-SW01	Speed Test	796Mbps
PCM-SITEA-SW01	Jitter Test	0.059ms
PCM-SITEA-SW02	Ping Test	0.566ms 0 packet loss
PCM-SITEA-SW02	Speed Test	790Mbps
PCM-SITEA-SW02	Jitter Test	0.042ms
PCM-SITEB-SW01	Ping Test	0.686ms 0 packet loss
PCM-SITEB-SW01	Speed Test	860Mbps
PCM-SITEB-SW01	Jitter Test	0.45ms
PCM-SITEC-SW01	Ping Test	0.827ms 0 packet loss
PCM-SITEC-SW01	Speed Test	860Mbps
PCM-SITEC-SW01	Jitter Test	0.056ms
PCM-SITED-SW01	Ping Test	0.661ms

		0 packet loss
PCM-SITED-SW01	Speed Test	860Mbps
PCM-SITED-SW01	Jitter Test	0.048ms
PCM-SITEA-AP01	Ping Test	0.727ms 0 packet loss
PCM-SITEA-AP01	Speed Test	861Mbps
PCM-SITEA-AP01	Jitter Test	0.045ms
PCM-SITEA-AP02	Ping Test	0.624ms 0 packet loss
PCM-SITEA-AP02	Speed Test	856Mbps
PCM-SITEA-AP02	Jitter Test	0.045ms

Switch Results

Below are the screenshots taken from each switch/AP test

PCM-SITEA-SW01

PCM-SITEA-SW01 Ping Test

```

QEMU (PCM-SITEA-SW01)
root@PCM-SITEA-SW01:~# ping -c 4 192.168.199.40
PING 192.168.199.40 (192.168.199.40) 56(84) bytes of data.
64 bytes from 192.168.199.40: icmp_seq=1 ttl=64 time=0.462 ms
64 bytes from 192.168.199.40: icmp_seq=2 ttl=64 time=0.622 ms
64 bytes from 192.168.199.40: icmp_seq=3 ttl=64 time=0.575 ms
64 bytes from 192.168.199.40: icmp_seq=4 ttl=64 time=0.680 ms

--- 192.168.199.40 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.462/0.584/0.680/0.080 ms
root@PCM-SITEA-SW01:~# _

```

Figure 1 – screenshot of PCM-SITEA-SW01 ping test

PCM-SITEA-SW01 Speed Test

QEMU (PCM-SITEA-SW01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	97.6 MBytes	818 Mbits/sec	0	1.23 MBytes
[5]	1.00-2.00	sec	95.8 MBytes	803 Mbits/sec	0	1.62 MBytes
[5]	2.00-3.00	sec	92.4 MBytes	775 Mbits/sec	0	2.33 MBytes
[5]	3.00-4.00	sec	98.1 MBytes	822 Mbits/sec	0	2.33 MBytes
[5]	4.00-5.00	sec	96.4 MBytes	809 Mbits/sec	0	2.33 MBytes
[5]	5.00-6.00	sec	96.8 MBytes	811 Mbits/sec	0	2.33 MBytes
[5]	6.00-7.00	sec	95.4 MBytes	800 Mbits/sec	0	2.46 MBytes
[5]	7.00-8.00	sec	94.9 MBytes	796 Mbits/sec	0	2.46 MBytes
[5]	8.00-9.00	sec	97.6 MBytes	819 Mbits/sec	0	2.46 MBytes
[5]	9.00-10.00	sec	95.9 MBytes	805 Mbits/sec	0	2.46 MBytes
[5]	10.00-11.00	sec	96.2 MBytes	807 Mbits/sec	0	2.46 MBytes
[5]	11.00-12.00	sec	96.6 MBytes	812 Mbits/sec	0	2.46 MBytes
[5]	12.00-13.00	sec	96.5 MBytes	810 Mbits/sec	0	2.87 MBytes
[5]	13.00-14.00	sec	97.8 MBytes	819 Mbits/sec	0	2.87 MBytes
[5]	14.00-15.00	sec	95.6 MBytes	803 Mbits/sec	0	2.87 MBytes
[5]	15.00-16.00	sec	96.6 MBytes	810 Mbits/sec	0	2.87 MBytes
[5]	16.00-17.00	sec	95.1 MBytes	798 Mbits/sec	0	2.87 MBytes
[5]	17.00-18.00	sec	91.6 MBytes	768 Mbits/sec	0	3.09 MBytes
[5]	18.00-19.00	sec	95.6 MBytes	802 Mbits/sec	0	3.09 MBytes
[5]	19.00-20.00	sec	88.8 MBytes	745 Mbits/sec	0	4.04 MBytes
[5]	20.00-21.00	sec	92.5 MBytes	776 Mbits/sec	0	4.04 MBytes
[5]	21.00-22.00	sec	90.1 MBytes	756 Mbits/sec	0	4.04 MBytes
[5]	22.00-23.00	sec	94.0 MBytes	788 Mbits/sec	0	4.04 MBytes
[5]	23.00-24.00	sec	94.9 MBytes	796 Mbits/sec	0	4.04 MBytes
[5]	24.00-25.00	sec	95.8 MBytes	803 Mbits/sec	0	4.04 MBytes
[5]	25.00-26.00	sec	95.6 MBytes	802 Mbits/sec	0	4.04 MBytes
[5]	26.00-27.00	sec	96.4 MBytes	808 Mbits/sec	0	4.04 MBytes
[5]	27.00-28.00	sec	94.4 MBytes	792 Mbits/sec	0	4.04 MBytes
[5]	28.00-29.00	sec	96.2 MBytes	807 Mbits/sec	0	4.04 MBytes
[5]	29.00-30.00	sec	96.9 MBytes	812 Mbits/sec	0	4.04 MBytes
[5]	30.00-31.00	sec	93.4 MBytes	783 Mbits/sec	0	4.04 MBytes
[5]	31.00-32.00	sec	94.5 MBytes	793 Mbits/sec	0	4.04 MBytes
[5]	32.00-33.00	sec	90.0 MBytes	755 Mbits/sec	0	4.04 MBytes
[5]	33.00-34.00	sec	95.0 MBytes	797 Mbits/sec	0	4.04 MBytes
[5]	34.00-35.00	sec	94.9 MBytes	796 Mbits/sec	0	4.04 MBytes
[5]	35.00-36.00	sec	90.4 MBytes	758 Mbits/sec	0	4.04 MBytes
[5]	36.00-37.00	sec	91.5 MBytes	768 Mbits/sec	0	4.04 MBytes
[5]	37.00-38.00	sec	97.0 MBytes	814 Mbits/sec	0	4.04 MBytes
[5]	38.00-39.00	sec	96.4 MBytes	808 Mbits/sec	0	4.04 MBytes
[5]	39.00-40.00	sec	97.8 MBytes	820 Mbits/sec	0	4.04 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	3.71 GBytes	797 Mbits/sec	0	sender
[5]	0.00-40.02	sec	3.71 GBytes	796 Mbits/sec		receiver

iperf Done.
root@PCM-SITEA-SW01:~#

Figure 2 – screenshot of PCM-SITEA-SW01 speed test

PCM-SITEA-SW01 Jitter Test

```

QEMU (PCM-SITEA-SW01)
[ 5] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00     sec 47.5 MBytes 398 Mbits/sec 35577
[ 5] 1.00-2.00     sec 45.2 MBytes 379 Mbits/sec 33836
[ 5] 2.00-3.00     sec 46.9 MBytes 393 Mbits/sec 35137
[ 5] 3.00-4.00     sec 44.2 MBytes 356 Mbits/sec 33132
[ 5] 4.00-5.00     sec 44.8 MBytes 394 Mbits/sec 33584
[ 5] 5.00-6.00     sec 43.0 MBytes 361 Mbits/sec 32216
[ 5] 6.00-7.00     sec 42.6 MBytes 358 Mbits/sec 31922
[ 5] 7.00-8.00     sec 45.2 MBytes 379 Mbits/sec 33848
[ 5] 8.00-9.00     sec 44.2 MBytes 371 Mbits/sec 33140
[ 5] 9.00-10.00    sec 40.4 MBytes 339 Mbits/sec 30228
[ 5] 10.00-11.00   sec 48.4 MBytes 406 Mbits/sec 36269
[ 5] 11.00-12.00   sec 43.1 MBytes 361 Mbits/sec 32305
[ 5] 12.00-13.00   sec 47.6 MBytes 400 Mbits/sec 35664
[ 5] 13.00-14.00   sec 47.0 MBytes 394 Mbits/sec 35174
[ 5] 14.00-15.00   sec 44.8 MBytes 375 Mbits/sec 33518
[ 5] 15.00-16.00   sec 49.8 MBytes 418 Mbits/sec 37321
[ 5] 16.00-17.00   sec 45.2 MBytes 379 Mbits/sec 33831
[ 5] 17.00-18.00   sec 44.6 MBytes 375 Mbits/sec 33440
[ 5] 18.00-19.00   sec 43.6 MBytes 366 Mbits/sec 32655
[ 5] 19.00-20.00   sec 45.1 MBytes 378 Mbits/sec 33752
[ 5] 20.00-21.00   sec 48.3 MBytes 405 Mbits/sec 36184
[ 5] 21.00-22.00   sec 44.1 MBytes 370 Mbits/sec 33038
[ 5] 22.00-23.00   sec 46.8 MBytes 393 Mbits/sec 35044
[ 5] 23.00-24.00   sec 49.8 MBytes 417 Mbits/sec 37290
[ 5] 24.00-25.00   sec 44.5 MBytes 374 Mbits/sec 33343
[ 5] 25.00-26.00   sec 51.1 MBytes 428 Mbits/sec 38276
[ 5] 26.00-27.00   sec 42.2 MBytes 354 Mbits/sec 31576
[ 5] 27.00-28.00   sec 42.5 MBytes 356 Mbits/sec 31815
[ 5] 28.00-29.00   sec 47.5 MBytes 398 Mbits/sec 35590
[ 5] 29.00-30.00   sec 41.6 MBytes 350 Mbits/sec 31187
[ 5] 30.00-31.00   sec 42.9 MBytes 360 Mbits/sec 32148
[ 5] 31.00-32.00   sec 45.2 MBytes 379 Mbits/sec 33859
[ 5] 32.00-33.00   sec 47.9 MBytes 402 Mbits/sec 35880
[ 5] 33.00-34.00   sec 49.6 MBytes 416 Mbits/sec 37118
[ 5] 34.00-35.00   sec 46.1 MBytes 387 Mbits/sec 34548
[ 5] 35.00-36.00   sec 47.4 MBytes 397 Mbits/sec 35472
[ 5] 36.00-37.00   sec 46.6 MBytes 391 Mbits/sec 34878
[ 5] 37.00-38.00   sec 49.4 MBytes 415 Mbits/sec 37026
[ 5] 38.00-39.00   sec 45.5 MBytes 382 Mbits/sec 34083
[ 5] 39.00-40.00   sec 50.7 MBytes 425 Mbits/sec 37996
[ 5] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00     sec 1.79 GBytes 384 Mbits/sec 0.000 ms 0/1372900 (0%) sender
[ 5] 0.00-40.00     sec 1.54 GBytes 331 Mbits/sec 0.059 ms 190043/1372900 (14%) receiver

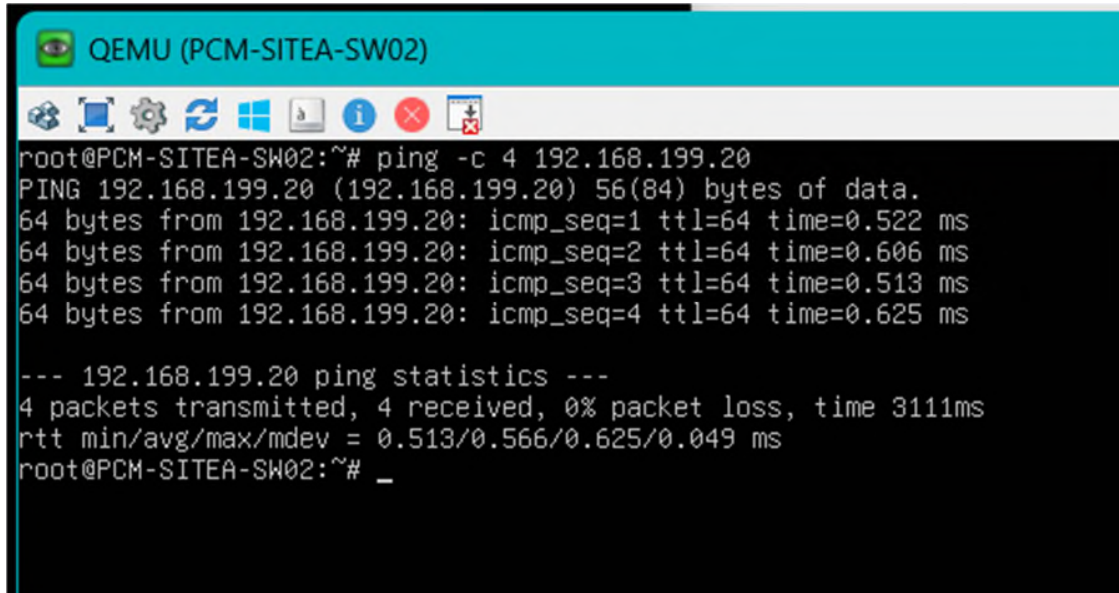
iperf Done.
root@PCM-SITEA-SW01:~#

```

Figure 3 – screenshot of PCM-SITEA-SW01 jitter test

PCM-SITEA-SW02

PCM-SITEA-SW02 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEA-SW02)". The terminal displays the output of a ping command executed from the root user at the PCM-SITEA-SW02 host. The command is "ping -c 4 192.168.199.20". The output shows four successful ping requests, each with a 64-byte payload, a TTL of 64, and a response time between 0.513 ms and 0.625 ms. The statistics section indicates that 4 packets were transmitted and received with 0% packet loss, and the total time for the test was 3111 ms. The round-trip time (RTT) statistics are: min/avg/max/mdev = 0.513/0.566/0.625/0.049 ms.

```
root@PCM-SITEA-SW02:~# ping -c 4 192.168.199.20
PING 192.168.199.20 (192.168.199.20) 56(84) bytes of data.
64 bytes from 192.168.199.20: icmp_seq=1 ttl=64 time=0.522 ms
64 bytes from 192.168.199.20: icmp_seq=2 ttl=64 time=0.606 ms
64 bytes from 192.168.199.20: icmp_seq=3 ttl=64 time=0.513 ms
64 bytes from 192.168.199.20: icmp_seq=4 ttl=64 time=0.625 ms

--- 192.168.199.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.513/0.566/0.625/0.049 ms
root@PCM-SITEA-SW02:~# _
```

Figure 4 – screenshot of PCM-SITEA-SW02 ping test

PCM-SITEA-SW02 Speed Test

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	83.5 MBytes	700 Mbits/sec	0	1.37 MBytes
[5]	1.00-2.00	sec	94.2 MBytes	791 Mbits/sec	0	1.71 MBytes
[5]	2.00-3.00	sec	95.9 MBytes	805 Mbits/sec	0	1.80 MBytes
[5]	3.00-4.00	sec	97.2 MBytes	816 Mbits/sec	0	1.80 MBytes
[5]	4.00-5.00	sec	79.0 MBytes	662 Mbits/sec	0	1.98 MBytes
[5]	5.00-6.00	sec	95.0 MBytes	798 Mbits/sec	0	1.98 MBytes
[5]	6.00-7.00	sec	96.0 MBytes	805 Mbits/sec	0	1.98 MBytes
[5]	7.00-8.00	sec	94.2 MBytes	790 Mbits/sec	0	2.08 MBytes
[5]	8.00-9.00	sec	89.4 MBytes	751 Mbits/sec	0	3.21 MBytes
[5]	9.00-10.00	sec	92.4 MBytes	775 Mbits/sec	0	3.21 MBytes
[5]	10.00-11.00	sec	97.9 MBytes	821 Mbits/sec	0	3.21 MBytes
[5]	11.00-12.00	sec	94.9 MBytes	796 Mbits/sec	0	3.21 MBytes
[5]	12.00-13.00	sec	94.9 MBytes	796 Mbits/sec	0	3.21 MBytes
[5]	13.00-14.00	sec	95.5 MBytes	800 Mbits/sec	0	3.21 MBytes
[5]	14.00-15.00	sec	96.5 MBytes	810 Mbits/sec	0	3.21 MBytes
[5]	15.00-16.00	sec	96.5 MBytes	810 Mbits/sec	0	3.21 MBytes
[5]	16.00-17.00	sec	96.9 MBytes	813 Mbits/sec	0	3.21 MBytes
[5]	17.00-18.00	sec	96.2 MBytes	807 Mbits/sec	0	3.21 MBytes
[5]	18.00-19.00	sec	95.4 MBytes	801 Mbits/sec	0	3.21 MBytes
[5]	19.00-20.00	sec	94.8 MBytes	794 Mbits/sec	0	3.21 MBytes
[5]	20.00-21.00	sec	94.1 MBytes	791 Mbits/sec	0	3.21 MBytes
[5]	21.00-22.00	sec	92.9 MBytes	778 Mbits/sec	0	3.21 MBytes
[5]	22.00-23.00	sec	93.4 MBytes	783 Mbits/sec	0	3.21 MBytes
[5]	23.00-24.00	sec	94.5 MBytes	793 Mbits/sec	0	3.21 MBytes
[5]	24.00-25.00	sec	95.4 MBytes	800 Mbits/sec	0	3.21 MBytes
[5]	25.00-26.00	sec	94.8 MBytes	795 Mbits/sec	0	4.89 MBytes
[5]	26.00-27.00	sec	98.5 MBytes	826 Mbits/sec	0	4.89 MBytes
[5]	27.00-28.00	sec	96.5 MBytes	809 Mbits/sec	0	4.89 MBytes
[5]	28.00-29.00	sec	95.0 MBytes	797 Mbits/sec	0	4.89 MBytes
[5]	29.00-30.00	sec	96.5 MBytes	810 Mbits/sec	0	4.89 MBytes
[5]	30.00-31.00	sec	96.5 MBytes	809 Mbits/sec	0	4.89 MBytes
[5]	31.00-32.00	sec	97.2 MBytes	815 Mbits/sec	0	4.89 MBytes
[5]	32.00-33.00	sec	95.1 MBytes	799 Mbits/sec	0	4.89 MBytes
[5]	33.00-34.00	sec	96.8 MBytes	811 Mbits/sec	0	4.89 MBytes
[5]	34.00-35.00	sec	95.5 MBytes	802 Mbits/sec	0	4.89 MBytes
[5]	35.00-36.00	sec	93.6 MBytes	785 Mbits/sec	0	4.89 MBytes
[5]	36.00-37.00	sec	95.1 MBytes	798 Mbits/sec	0	4.89 MBytes
[5]	37.00-38.00	sec	92.5 MBytes	776 Mbits/sec	0	4.89 MBytes
[5]	38.00-39.00	sec	91.8 MBytes	769 Mbits/sec	0	4.89 MBytes
[5]	39.00-40.00	sec	86.9 MBytes	729 Mbits/sec	0	4.89 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	3.68 GBytes	790 Mbits/sec	0	sender
[5]	0.00-40.03	sec	3.68 GBytes	790 Mbits/sec	0	receiver

iperf Done.
root@PCM-SITEA-SW02:~#

Figure 5 – screenshot of PCM-SITEA-SW02 speed test

PCM-SITEA-SW02 Jitter Test

```

QEMU (PCM-SITEA-SW02)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  47.3 MBytes  396 Mbits/sec  35419
[ 5]  1.00-2.00    sec  46.5 MBytes  391 Mbits/sec  34856
[ 5]  2.00-3.00    sec  46.8 MBytes  392 Mbits/sec  35053
[ 5]  3.00-4.00    sec  44.7 MBytes  375 Mbits/sec  33470
[ 5]  4.00-5.00    sec  48.3 MBytes  405 Mbits/sec  36154
[ 5]  5.00-6.00    sec  47.2 MBytes  396 Mbits/sec  35340
[ 5]  6.00-7.00    sec  42.8 MBytes  359 Mbits/sec  32069
[ 5]  7.00-8.00    sec  45.6 MBytes  383 Mbits/sec  34127
[ 5]  8.00-9.00    sec  37.2 MBytes  312 Mbits/sec  27884
[ 5]  9.00-10.00   sec  46.8 MBytes  393 Mbits/sec  35060
[ 5] 10.00-11.00   sec  47.7 MBytes  400 Mbits/sec  35695
[ 5] 11.00-12.00   sec  45.9 MBytes  385 Mbits/sec  34402
[ 5] 12.00-13.00   sec  45.0 MBytes  378 Mbits/sec  33717
[ 5] 13.00-14.00   sec  45.9 MBytes  385 Mbits/sec  34409
[ 5] 14.00-15.00   sec  41.1 MBytes  345 Mbits/sec  30761
[ 5] 15.00-16.00   sec  46.8 MBytes  392 Mbits/sec  35035
[ 5] 16.00-17.00   sec  44.6 MBytes  374 Mbits/sec  33418
[ 5] 17.00-18.00   sec  47.1 MBytes  395 Mbits/sec  35284
[ 5] 18.00-19.00   sec  45.2 MBytes  379 Mbits/sec  33862
[ 5] 19.00-20.00   sec  44.8 MBytes  375 Mbits/sec  33558
[ 5] 20.00-21.00   sec  48.2 MBytes  404 Mbits/sec  36076
[ 5] 21.00-22.00   sec  45.1 MBytes  379 Mbits/sec  33796
[ 5] 22.00-23.00   sec  46.6 MBytes  390 Mbits/sec  34898
[ 5] 23.00-24.00   sec  43.8 MBytes  368 Mbits/sec  32801
[ 5] 24.00-25.00   sec  46.5 MBytes  389 Mbits/sec  34801
[ 5] 25.00-26.00   sec  45.8 MBytes  384 Mbits/sec  34323
[ 5] 26.00-27.00   sec  43.5 MBytes  365 Mbits/sec  32573
[ 5] 27.00-28.00   sec  48.3 MBytes  406 Mbits/sec  36211
[ 5] 28.00-29.00   sec  47.1 MBytes  395 Mbits/sec  35273
[ 5] 29.00-30.00   sec  47.6 MBytes  399 Mbits/sec  35620
[ 5] 30.00-31.00   sec  43.3 MBytes  363 Mbits/sec  32411
[ 5] 31.00-32.00   sec  42.9 MBytes  361 Mbits/sec  32121
[ 5] 32.00-33.00   sec  47.6 MBytes  399 Mbits/sec  35638
[ 5] 33.00-34.00   sec  44.6 MBytes  374 Mbits/sec  33397
[ 5] 34.00-35.00   sec  45.4 MBytes  380 Mbits/sec  33973
[ 5] 35.00-36.00   sec  45.2 MBytes  379 Mbits/sec  33844
[ 5] 36.00-37.00   sec  45.8 MBytes  384 Mbits/sec  34306
[ 5] 37.00-38.00   sec  43.4 MBytes  363 Mbits/sec  32472
[ 5] 38.00-39.00   sec  47.4 MBytes  398 Mbits/sec  35497
[ 5] 39.00-40.00   sec  52.6 MBytes  441 Mbits/sec  39416
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00    sec  1.78 GBytes  383 Mbits/sec  0.000 ms    0/1369020 (0%) sender
[ 5]  0.00-40.07    sec  1.58 GBytes  338 Mbits/sec  0.042 ms   159242/1369020 (12%) receiver

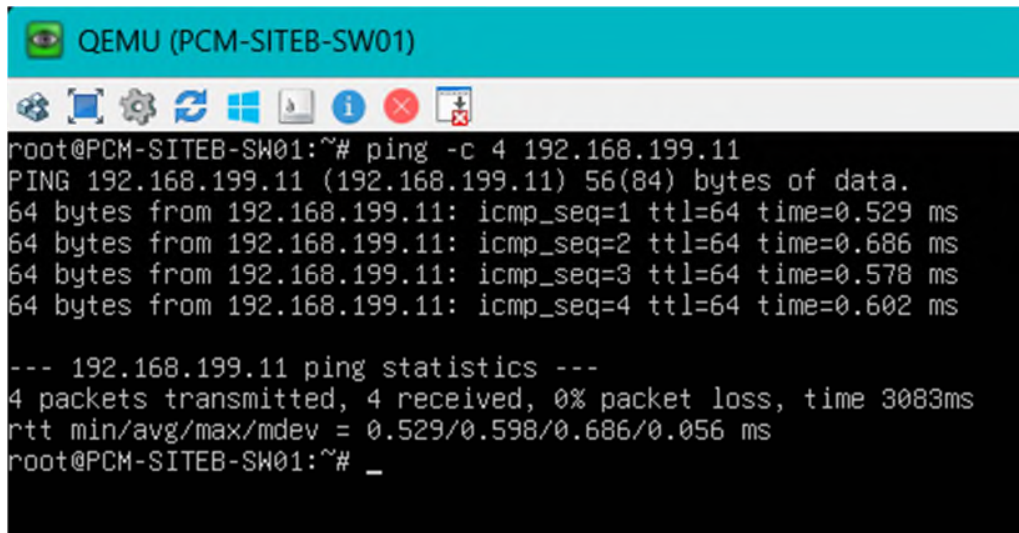
iperf Done.
root@PCM-SITEA-SW02:~#

```

Figure 6 – screenshot of PCM-SITEA-SW02 jitter test

PCM-SITEB-SW01

PCM-SITEB-SW01 Ping Test



```
QEMU (PCM-SITEB-SW01)
root@PCM-SITEB-SW01:~# ping -c 4 192.168.199.11
PING 192.168.199.11 (192.168.199.11) 56(84) bytes of data.
64 bytes from 192.168.199.11: icmp_seq=1 ttl=64 time=0.529 ms
64 bytes from 192.168.199.11: icmp_seq=2 ttl=64 time=0.686 ms
64 bytes from 192.168.199.11: icmp_seq=3 ttl=64 time=0.578 ms
64 bytes from 192.168.199.11: icmp_seq=4 ttl=64 time=0.602 ms

--- 192.168.199.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3083ms
rtt min/avg/max/mdev = 0.529/0.598/0.686/0.056 ms
root@PCM-SITEB-SW01:~# _
```

Figure 7 – screenshot of PCM-SITEB-SW01 ping test

PCM-SITEB-SW01 Speed Test

QEMU (PCM-SITEB-SW01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	103 MBytes	866 Mbits/sec	0	689 KBytes
[5]	1.00-2.00	sec	101 MBytes	845 Mbits/sec	0	689 KBytes
[5]	2.00-3.00	sec	106 MBytes	886 Mbits/sec	0	689 KBytes
[5]	3.00-4.00	sec	101 MBytes	846 Mbits/sec	0	725 KBytes
[5]	4.00-5.00	sec	105 MBytes	879 Mbits/sec	0	725 KBytes
[5]	5.00-6.00	sec	98.8 MBytes	828 Mbits/sec	0	765 KBytes
[5]	6.00-7.00	sec	104 MBytes	876 Mbits/sec	0	765 KBytes
[5]	7.00-8.00	sec	105 MBytes	878 Mbits/sec	0	765 KBytes
[5]	8.00-9.00	sec	102 MBytes	852 Mbits/sec	0	803 KBytes
[5]	9.00-10.00	sec	103 MBytes	866 Mbits/sec	0	803 KBytes
[5]	10.00-11.00	sec	102 MBytes	854 Mbits/sec	0	884 KBytes
[5]	11.00-12.00	sec	103 MBytes	864 Mbits/sec	0	884 KBytes
[5]	12.00-13.00	sec	103 MBytes	866 Mbits/sec	0	884 KBytes
[5]	13.00-14.00	sec	101 MBytes	846 Mbits/sec	0	1.57 MBytes
[5]	14.00-15.00	sec	104 MBytes	874 Mbits/sec	0	1.57 MBytes
[5]	15.00-16.00	sec	101 MBytes	850 Mbits/sec	0	1.57 MBytes
[5]	16.00-17.00	sec	104 MBytes	872 Mbits/sec	0	1.57 MBytes
[5]	17.00-18.00	sec	103 MBytes	864 Mbits/sec	0	1.57 MBytes
[5]	18.00-19.00	sec	101 MBytes	845 Mbits/sec	0	2.47 MBytes
[5]	19.00-20.00	sec	102 MBytes	852 Mbits/sec	0	2.47 MBytes
[5]	20.00-21.00	sec	97.6 MBytes	818 Mbits/sec	0	2.47 MBytes
[5]	21.00-22.00	sec	103 MBytes	864 Mbits/sec	0	2.47 MBytes
[5]	22.00-23.00	sec	102 MBytes	854 Mbits/sec	0	2.47 MBytes
[5]	23.00-24.00	sec	100 MBytes	842 Mbits/sec	0	2.47 MBytes
[5]	24.00-25.00	sec	103 MBytes	867 Mbits/sec	0	2.47 MBytes
[5]	25.00-26.00	sec	102 MBytes	857 Mbits/sec	0	2.47 MBytes
[5]	26.00-27.00	sec	103 MBytes	864 Mbits/sec	0	2.47 MBytes
[5]	27.00-28.00	sec	105 MBytes	878 Mbits/sec	0	2.47 MBytes
[5]	28.00-29.00	sec	102 MBytes	854 Mbits/sec	0	2.47 MBytes
[5]	29.00-30.00	sec	104 MBytes	877 Mbits/sec	0	2.47 MBytes
[5]	30.00-31.00	sec	104 MBytes	872 Mbits/sec	0	3.78 MBytes
[5]	31.00-32.00	sec	103 MBytes	864 Mbits/sec	0	3.78 MBytes
[5]	32.00-33.00	sec	102 MBytes	855 Mbits/sec	0	3.78 MBytes
[5]	33.00-34.00	sec	98.9 MBytes	827 Mbits/sec	0	3.78 MBytes
[5]	34.00-35.00	sec	102 MBytes	859 Mbits/sec	0	3.78 MBytes
[5]	35.00-36.01	sec	103 MBytes	853 Mbits/sec	0	3.78 MBytes
[5]	36.01-37.00	sec	102 MBytes	865 Mbits/sec	0	3.78 MBytes
[5]	37.00-38.00	sec	105 MBytes	878 Mbits/sec	0	3.78 MBytes
[5]	38.00-39.00	sec	102 MBytes	852 Mbits/sec	0	3.78 MBytes
[5]	39.00-40.00	sec	104 MBytes	875 Mbits/sec	0	3.78 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	860 Mbits/sec	0	sender
[5]	0.00-40.01	sec	4.00 GBytes	860 Mbits/sec		receiver

iperf Done.
root@PCM-SITEB-SW01:~#

Figure 8 – screenshot of PCM-SITEB-SW01 speed test

PCM-SITEB-SW01 Jitter Test

QEMU (PCM-SITEB-SW01)

[ID]	Interval		Transfer	Bitrate	Total	Datagrams
[5]	0.00-1.00	sec	46.2 MBytes	388 Mb/s	34623	
[5]	1.00-2.00	sec	47.0 MBytes	394 Mb/s	35210	
[5]	2.00-3.00	sec	45.6 MBytes	382 Mb/s	34123	
[5]	3.00-4.00	sec	42.4 MBytes	355 Mb/s	31778	
[5]	4.00-5.00	sec	46.1 MBytes	387 Mb/s	34493	
[5]	5.00-6.00	sec	44.3 MBytes	372 Mb/s	33170	
[5]	6.00-7.00	sec	45.6 MBytes	382 Mb/s	34132	
[5]	7.00-8.00	sec	45.7 MBytes	384 Mb/s	34245	
[5]	8.00-9.00	sec	45.3 MBytes	380 Mb/s	33917	
[5]	9.00-10.00	sec	43.7 MBytes	367 Mb/s	32745	
[5]	10.00-11.00	sec	43.6 MBytes	366 Mb/s	32652	
[5]	11.00-12.00	sec	45.6 MBytes	383 Mb/s	34161	
[5]	12.00-13.00	sec	44.6 MBytes	374 Mb/s	33421	
[5]	13.00-14.00	sec	47.1 MBytes	395 Mb/s	35257	
[5]	14.00-15.00	sec	47.9 MBytes	402 Mb/s	35884	
[5]	15.00-16.00	sec	47.6 MBytes	398 Mb/s	35623	
[5]	16.00-17.00	sec	44.7 MBytes	375 Mb/s	33449	
[5]	17.00-18.00	sec	42.6 MBytes	357 Mb/s	31919	
[5]	18.00-19.00	sec	44.1 MBytes	370 Mb/s	33044	
[5]	19.00-20.00	sec	44.9 MBytes	376 Mb/s	33612	
[5]	20.00-21.00	sec	47.6 MBytes	399 Mb/s	35665	
[5]	21.00-22.00	sec	46.9 MBytes	393 Mb/s	35094	
[5]	22.00-23.00	sec	44.1 MBytes	370 Mb/s	33000	
[5]	23.00-24.00	sec	45.1 MBytes	378 Mb/s	33780	
[5]	24.00-25.00	sec	44.3 MBytes	372 Mb/s	33180	
[5]	25.00-26.00	sec	44.9 MBytes	377 Mb/s	33639	
[5]	26.00-27.00	sec	45.5 MBytes	382 Mb/s	34065	
[5]	27.00-28.00	sec	45.1 MBytes	378 Mb/s	33747	
[5]	28.00-29.00	sec	44.8 MBytes	376 Mb/s	33534	
[5]	29.00-30.00	sec	45.8 MBytes	384 Mb/s	34289	
[5]	30.00-31.00	sec	44.5 MBytes	373 Mb/s	33310	
[5]	31.00-32.00	sec	44.9 MBytes	376 Mb/s	33596	
[5]	32.00-33.00	sec	45.1 MBytes	379 Mb/s	33804	
[5]	33.00-34.00	sec	45.3 MBytes	380 Mb/s	33950	
[5]	34.00-35.00	sec	46.4 MBytes	390 Mb/s	34790	
[5]	35.00-36.00	sec	45.3 MBytes	380 Mb/s	33910	
[5]	36.00-37.00	sec	43.3 MBytes	363 Mb/s	32454	
[5]	37.00-38.00	sec	45.1 MBytes	378 Mb/s	33766	
[5]	38.00-39.00	sec	46.0 MBytes	386 Mb/s	34423	
[5]	39.00-40.00	sec	44.8 MBytes	375 Mb/s	33541	

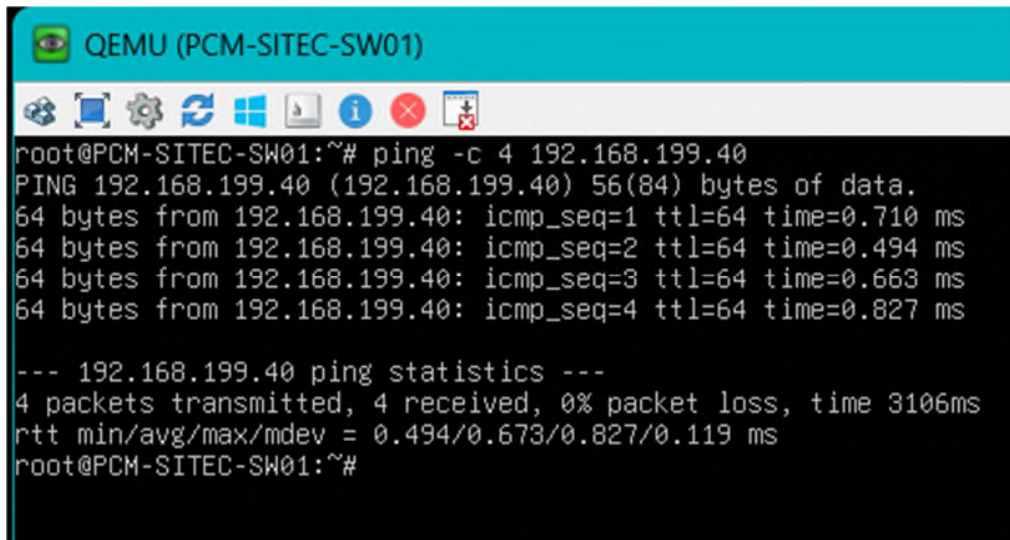
[ID]	Interval		Transfer	Bitrate	Jitter	Lost/Total Datagrams
[5]	0.00-40.00	sec	1.77 GBytes	379 Mb/s	0.000 ms	0/1354995 (0%) sender
[5]	0.00-40.00	sec	1.76 GBytes	378 Mb/s	0.045 ms	5747/1354995 (0.42%) receiver

iperf Done.
root@PCM-SITEB-SW01:~#

Figure 9 – screenshot of PCM-SITEB-SW01 jitter test

PCM-SITEC-SW01

PCM-SITEC-SW01 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEC-SW01)". The terminal displays the output of a ping command executed from the root user at the PCM-SITEC-SW01 host. The command is "ping -c 4 192.168.199.40". The output shows four successful ping requests, each with a 64-byte payload, a TTL of 64, and a response time between 0.494 ms and 0.827 ms. The statistics show 4 packets transmitted, 4 received, 0% packet loss, and a total time of 3106ms. The round-trip time (RTT) statistics are: min/avg/max/mdev = 0.494/0.673/0.827/0.119 ms.

```
root@PCM-SITEC-SW01:~# ping -c 4 192.168.199.40
PING 192.168.199.40 (192.168.199.40) 56(84) bytes of data.
64 bytes from 192.168.199.40: icmp_seq=1 ttl=64 time=0.710 ms
64 bytes from 192.168.199.40: icmp_seq=2 ttl=64 time=0.494 ms
64 bytes from 192.168.199.40: icmp_seq=3 ttl=64 time=0.663 ms
64 bytes from 192.168.199.40: icmp_seq=4 ttl=64 time=0.827 ms

--- 192.168.199.40 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3106ms
rtt min/avg/max/mdev = 0.494/0.673/0.827/0.119 ms
root@PCM-SITEC-SW01:~#
```

Figure 10 – screenshot of PCM-SITEC-SW01 ping test

PCM-SITEC-SW01 Speed Test

```

QEMU (PCM-SITEC-SW01)
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00      sec    103 MBytes    864 Mbits/sec    0    471 KBytes
[ 5]  1.00-2.00      sec    106 MBytes    886 Mbits/sec    0    494 KBytes
[ 5]  2.00-3.00      sec    106 MBytes    887 Mbits/sec    0    515 KBytes
[ 5]  3.00-4.00      sec    102 MBytes    854 Mbits/sec    0    568 KBytes
[ 5]  4.00-5.00      sec    102 MBytes    854 Mbits/sec    0    631 KBytes
[ 5]  5.00-6.00      sec   99.6 MBytes    836 Mbits/sec    0    701 KBytes
[ 5]  6.00-7.00      sec    105 MBytes    879 Mbits/sec    0    701 KBytes
[ 5]  7.00-8.00      sec    103 MBytes    863 Mbits/sec    0    701 KBytes
[ 5]  8.00-9.00      sec   98.9 MBytes    829 Mbits/sec    0    701 KBytes
[ 5]  9.00-10.00     sec    101 MBytes    844 Mbits/sec    0    773 KBytes
[ 5] 10.00-11.00     sec    102 MBytes    855 Mbits/sec    0    831 KBytes
[ 5] 11.00-12.00     sec    100 MBytes    841 Mbits/sec    0    831 KBytes
[ 5] 12.00-13.00     sec    103 MBytes    864 Mbits/sec    0    831 KBytes
[ 5] 13.00-14.00     sec    102 MBytes    852 Mbits/sec    0    1.23 MBytes
[ 5] 14.00-15.00     sec    102 MBytes    852 Mbits/sec    0    1.23 MBytes
[ 5] 15.00-16.00     sec    100 MBytes    839 Mbits/sec    0    1.87 MBytes
[ 5] 16.00-17.00     sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 17.00-18.00     sec    103 MBytes    862 Mbits/sec    0    1.87 MBytes
[ 5] 18.00-19.00     sec    101 MBytes    849 Mbits/sec    0    1.87 MBytes
[ 5] 19.00-20.00     sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 20.00-21.00     sec    101 MBytes    850 Mbits/sec    0    1.87 MBytes
[ 5] 21.00-22.00     sec    104 MBytes    875 Mbits/sec    0    1.87 MBytes
[ 5] 22.00-23.00     sec    104 MBytes    872 Mbits/sec    0    1.87 MBytes
[ 5] 23.00-24.00     sec    100 MBytes    841 Mbits/sec    0    1.87 MBytes
[ 5] 24.00-25.00     sec    105 MBytes    884 Mbits/sec    0    1.87 MBytes
[ 5] 25.00-26.00     sec    103 MBytes    862 Mbits/sec    0    1.87 MBytes
[ 5] 26.00-27.00     sec    104 MBytes    873 Mbits/sec    0    1.87 MBytes
[ 5] 27.00-28.00     sec    102 MBytes    854 Mbits/sec    0    1.87 MBytes
[ 5] 28.00-29.00     sec    104 MBytes    869 Mbits/sec    0    2.84 MBytes
[ 5] 29.00-30.00     sec    103 MBytes    862 Mbits/sec    0    2.84 MBytes
[ 5] 30.00-31.00     sec    104 MBytes    874 Mbits/sec    0    2.84 MBytes
[ 5] 31.00-32.00     sec    103 MBytes    861 Mbits/sec    0    2.84 MBytes
[ 5] 32.00-33.00     sec    104 MBytes    868 Mbits/sec    0    2.84 MBytes
[ 5] 33.00-34.00     sec    102 MBytes    856 Mbits/sec    0    2.84 MBytes
[ 5] 34.00-35.00     sec    103 MBytes    865 Mbits/sec    0    2.84 MBytes
[ 5] 35.00-36.00     sec    104 MBytes    874 Mbits/sec    0    2.84 MBytes
[ 5] 36.00-37.00     sec    103 MBytes    864 Mbits/sec    0    2.84 MBytes
[ 5] 37.00-38.00     sec    103 MBytes    863 Mbits/sec    0    2.84 MBytes
[ 5] 38.00-39.00     sec    102 MBytes    853 Mbits/sec    0    2.84 MBytes
[ 5] 39.00-40.00     sec   99.5 MBytes    835 Mbits/sec    0    2.84 MBytes
-- -- -- -- --
[ ID] Interval      Transfer      Bitrate      Retr
[ 5]  0.00-40.00     sec    4.01 GBytes    860 Mbits/sec    0
[ 5]  0.00-40.01     sec    4.00 GBytes    860 Mbits/sec
sender
receiver

iperf Done.
root@PCM-SITEC-SW01:~#

```

Figure 11 – screenshot of PCM-SITEC-SW01 speed test

PCM-SITEC-SW01 Jitter Test

```

QEMU (PCM-SITEC-SW01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  45.9 MBytes  385 Mbits/sec  34391
[ 5]  1.00-2.00    sec  47.9 MBytes  402 Mbits/sec  35877
[ 5]  2.00-3.00    sec  44.3 MBytes  372 Mbits/sec  33186
[ 5]  3.00-4.00    sec  47.1 MBytes  395 Mbits/sec  35296
[ 5]  4.00-5.00    sec  45.8 MBytes  384 Mbits/sec  34335
[ 5]  5.00-6.00    sec  45.4 MBytes  381 Mbits/sec  34021
[ 5]  6.00-7.00    sec  44.9 MBytes  377 Mbits/sec  33649
[ 5]  7.00-8.00    sec  43.9 MBytes  368 Mbits/sec  32892
[ 5]  8.00-9.00    sec  43.8 MBytes  367 Mbits/sec  32806
[ 5]  9.00-10.00   sec  43.2 MBytes  362 Mbits/sec  32319
[ 5] 10.00-11.00   sec  41.7 MBytes  350 Mbits/sec  31230
[ 5] 11.00-12.00   sec  43.4 MBytes  364 Mbits/sec  32536
[ 5] 12.00-13.00   sec  41.6 MBytes  349 Mbits/sec  31155
[ 5] 13.00-14.00   sec  43.1 MBytes  362 Mbits/sec  32292
[ 5] 14.00-15.00   sec  46.0 MBytes  386 Mbits/sec  34454
[ 5] 15.00-16.00   sec  47.2 MBytes  396 Mbits/sec  35369
[ 5] 16.00-17.00   sec  45.3 MBytes  380 Mbits/sec  33899
[ 5] 17.00-18.00   sec  45.4 MBytes  381 Mbits/sec  33983
[ 5] 18.00-19.00   sec  42.6 MBytes  358 Mbits/sec  31938
[ 5] 19.00-20.00   sec  47.7 MBytes  401 Mbits/sec  35749
[ 5] 20.00-21.00   sec  48.0 MBytes  402 Mbits/sec  35957
[ 5] 21.00-22.00   sec  45.0 MBytes  378 Mbits/sec  33736
[ 5] 22.00-23.00   sec  46.8 MBytes  392 Mbits/sec  35059
[ 5] 23.00-24.00   sec  45.3 MBytes  380 Mbits/sec  33964
[ 5] 24.00-25.00   sec  46.6 MBytes  391 Mbits/sec  34918
[ 5] 25.00-26.00   sec  45.5 MBytes  382 Mbits/sec  34114
[ 5] 26.00-27.00   sec  42.2 MBytes  355 Mbits/sec  31644
[ 5] 27.00-28.00   sec  38.0 MBytes  318 Mbits/sec  28437
[ 5] 28.00-29.00   sec  44.0 MBytes  369 Mbits/sec  32924
[ 5] 29.00-30.00   sec  44.2 MBytes  371 Mbits/sec  33126
[ 5] 30.00-31.00   sec  46.2 MBytes  388 Mbits/sec  34614
[ 5] 31.00-32.00   sec  44.9 MBytes  376 Mbits/sec  33614
[ 5] 32.00-33.00   sec  45.0 MBytes  377 Mbits/sec  33699
[ 5] 33.00-34.00   sec  44.4 MBytes  372 Mbits/sec  33253
[ 5] 34.00-35.00   sec  45.7 MBytes  383 Mbits/sec  34252
[ 5] 35.00-36.00   sec  47.2 MBytes  396 Mbits/sec  35354
[ 5] 36.00-37.00   sec  46.7 MBytes  391 Mbits/sec  34941
[ 5] 37.00-38.00   sec  43.1 MBytes  362 Mbits/sec  32307
[ 5] 38.00-39.00   sec  46.2 MBytes  387 Mbits/sec  34581
[ 5] 39.00-40.00   sec  45.6 MBytes  382 Mbits/sec  34139

[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00    sec  1.75 GBytes  377 Mbits/sec  0.000 ms    0/1346010 (0%) sender
[ 5]  0.00-40.00    sec  1.75 GBytes  375 Mbits/sec  0.056 ms   5211/1346010 (0.39%) receiver

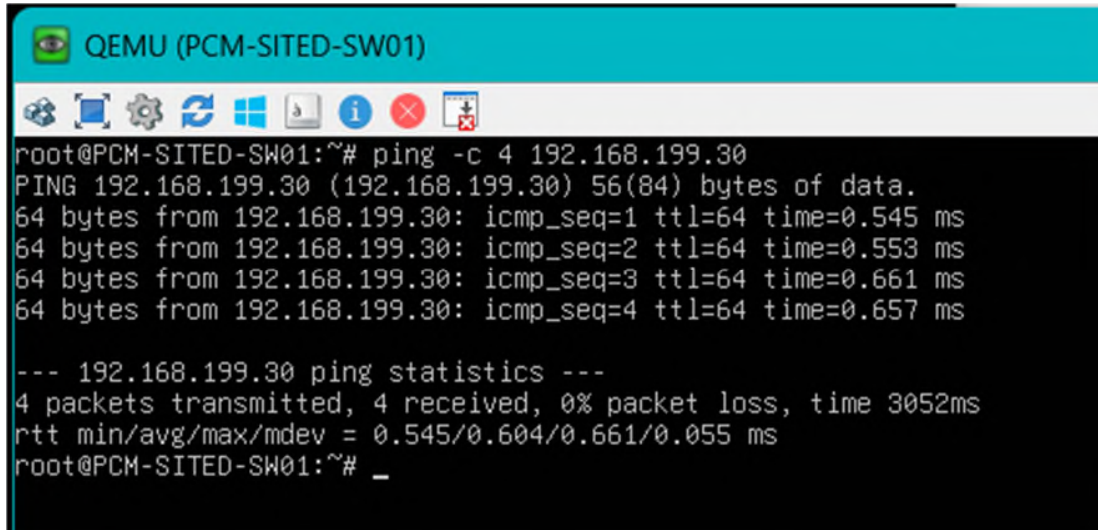
iperf Done.
root@PCM-SITEC-SW01:~#

```

Figure 12 – screenshot of PCM-SITEC-SW01 jitter test

PCM-SITED-SW01

PCM-SITED-SW01 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITED-SW01)". The terminal displays the output of a ping command executed from the root user at the PCM-SITED-SW01 host. The command is "ping -c 4 192.168.199.30". The output shows four successful ping requests, each with a 64-byte payload, a TTL of 64, and a response time between 0.545 ms and 0.661 ms. A summary line indicates that 4 packets were transmitted and received with 0% packet loss and a total time of 3052ms. The round-trip time (RTT) statistics are also shown: min/avg/max/mdev = 0.545/0.604/0.661/0.055 ms.

```
root@PCM-SITED-SW01:~# ping -c 4 192.168.199.30
PING 192.168.199.30 (192.168.199.30) 56(84) bytes of data.
64 bytes from 192.168.199.30: icmp_seq=1 ttl=64 time=0.545 ms
64 bytes from 192.168.199.30: icmp_seq=2 ttl=64 time=0.553 ms
64 bytes from 192.168.199.30: icmp_seq=3 ttl=64 time=0.661 ms
64 bytes from 192.168.199.30: icmp_seq=4 ttl=64 time=0.657 ms

--- 192.168.199.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3052ms
rtt min/avg/max/mdev = 0.545/0.604/0.661/0.055 ms
root@PCM-SITED-SW01:~# _
```

Figure 13 – screenshot of PCM-SITED-SW01 ping test

PCM-SITED-SW01 Speed Test

QEMU (PCM-SITED-SW01)						
[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	98.0 MBytes	822 Mb/s	0	757 KBytes
[5]	1.00-2.00	sec	100 MBytes	840 Mb/s	0	757 KBytes
[5]	2.00-3.00	sec	102 MBytes	854 Mb/s	0	795 KBytes
[5]	3.00-4.00	sec	101 MBytes	845 Mb/s	0	836 KBytes
[5]	4.00-5.00	sec	104 MBytes	873 Mb/s	0	836 KBytes
[5]	5.00-6.00	sec	103 MBytes	864 Mb/s	0	836 KBytes
[5]	6.00-7.00	sec	101 MBytes	849 Mb/s	0	836 KBytes
[5]	7.00-8.00	sec	104 MBytes	872 Mb/s	0	836 KBytes
[5]	8.00-9.00	sec	101 MBytes	851 Mb/s	0	836 KBytes
[5]	9.00-10.00	sec	104 MBytes	873 Mb/s	0	836 KBytes
[5]	10.00-11.00	sec	104 MBytes	875 Mb/s	0	836 KBytes
[5]	11.00-12.00	sec	101 MBytes	850 Mb/s	0	1.22 MBytes
[5]	12.00-13.00	sec	104 MBytes	873 Mb/s	0	1.22 MBytes
[5]	13.00-14.00	sec	100 MBytes	839 Mb/s	0	1.22 MBytes
[5]	14.00-15.00	sec	104 MBytes	874 Mb/s	0	1.22 MBytes
[5]	15.00-16.00	sec	104 MBytes	875 Mb/s	0	1.22 MBytes
[5]	16.00-17.00	sec	104 MBytes	875 Mb/s	0	1.22 MBytes
[5]	17.00-18.00	sec	102 MBytes	851 Mb/s	0	1.22 MBytes
[5]	18.00-19.00	sec	103 MBytes	863 Mb/s	0	1.22 MBytes
[5]	19.00-20.00	sec	103 MBytes	862 Mb/s	0	1.22 MBytes
[5]	20.00-21.00	sec	104 MBytes	872 Mb/s	0	1.22 MBytes
[5]	21.00-22.00	sec	104 MBytes	874 Mb/s	0	1.22 MBytes
[5]	22.00-23.00	sec	103 MBytes	863 Mb/s	0	1.22 MBytes
[5]	23.00-24.00	sec	104 MBytes	872 Mb/s	0	1.22 MBytes
[5]	24.00-25.00	sec	104 MBytes	873 Mb/s	0	1.22 MBytes
[5]	25.00-26.00	sec	103 MBytes	864 Mb/s	0	1.22 MBytes
[5]	26.00-27.00	sec	96.1 MBytes	806 Mb/s	0	1.22 MBytes
[5]	27.00-28.00	sec	101 MBytes	844 Mb/s	0	1.22 MBytes
[5]	28.00-29.00	sec	101 MBytes	848 Mb/s	0	1.83 MBytes
[5]	29.00-30.00	sec	103 MBytes	861 Mb/s	0	1.83 MBytes
[5]	30.00-31.00	sec	104 MBytes	874 Mb/s	0	1.83 MBytes
[5]	31.00-32.00	sec	103 MBytes	861 Mb/s	0	2.78 MBytes
[5]	32.00-33.00	sec	102 MBytes	852 Mb/s	0	2.78 MBytes
[5]	33.00-34.00	sec	104 MBytes	875 Mb/s	0	2.78 MBytes
[5]	34.00-35.00	sec	104 MBytes	871 Mb/s	0	2.78 MBytes
[5]	35.00-36.00	sec	104 MBytes	873 Mb/s	0	2.78 MBytes
[5]	36.00-37.00	sec	101 MBytes	846 Mb/s	0	2.78 MBytes
[5]	37.00-38.00	sec	104 MBytes	873 Mb/s	0	2.78 MBytes
[5]	38.00-39.00	sec	103 MBytes	863 Mb/s	0	2.78 MBytes
[5]	39.00-40.00	sec	104 MBytes	872 Mb/s	0	2.78 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	860 Mb/s	0	sender
[5]	0.00-40.01	sec	4.00 GBytes	860 Mb/s		receiver
iperf Done.						
root@PCM-SITED-SW01:~#						

Figure 14 – screenshot of PCM-SITED-SW01 speed test

PCM-SITED-SW01 Jitter Test

```

QEMU (PCM-SITED-SW01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  48.5 MBytes  407 Mb/s     36338
[ 5]  1.00-2.00    sec  42.1 MBytes  353 Mb/s     31496
[ 5]  2.00-3.00    sec  38.4 MBytes  322 Mb/s     28762
[ 5]  3.00-4.00    sec  46.1 MBytes  386 Mb/s     34503
[ 5]  4.00-5.00    sec  45.9 MBytes  385 Mb/s     34385
[ 5]  5.00-6.00    sec  44.1 MBytes  370 Mb/s     32997
[ 5]  6.00-7.00    sec  44.7 MBytes  375 Mb/s     33443
[ 5]  7.00-8.00    sec  45.6 MBytes  383 Mb/s     34164
[ 5]  8.00-9.00    sec  45.4 MBytes  381 Mb/s     34007
[ 5]  9.00-10.00   sec  47.3 MBytes  396 Mb/s     35418
[ 5] 10.00-11.00   sec  47.6 MBytes  400 Mb/s     35643
[ 5] 11.00-12.00   sec  46.0 MBytes  386 Mb/s     34479
[ 5] 12.00-13.00   sec  46.8 MBytes  392 Mb/s     35023
[ 5] 13.00-14.00   sec  44.4 MBytes  372 Mb/s     33254
[ 5] 14.00-15.00   sec  43.9 MBytes  368 Mb/s     32878
[ 5] 15.00-16.00   sec  42.8 MBytes  359 Mb/s     32057
[ 5] 16.00-17.00   sec  43.0 MBytes  361 Mb/s     32199
[ 5] 17.00-18.00   sec  41.7 MBytes  349 Mb/s     31215
[ 5] 18.00-19.00   sec  41.7 MBytes  350 Mb/s     31249
[ 5] 19.00-20.00   sec  47.9 MBytes  402 Mb/s     35896
[ 5] 20.00-21.00   sec  48.8 MBytes  409 Mb/s     36552
[ 5] 21.00-22.00   sec  44.3 MBytes  371 Mb/s     33163
[ 5] 22.00-23.00   sec  44.2 MBytes  370 Mb/s     33085
[ 5] 23.00-24.00   sec  44.3 MBytes  372 Mb/s     33216
[ 5] 24.00-25.00   sec  43.8 MBytes  367 Mb/s     32789
[ 5] 25.00-26.00   sec  44.7 MBytes  375 Mb/s     33481
[ 5] 26.00-27.00   sec  45.4 MBytes  381 Mb/s     33970
[ 5] 27.00-28.00   sec  46.8 MBytes  392 Mb/s     35022
[ 5] 28.00-29.00   sec  46.4 MBytes  389 Mb/s     34716
[ 5] 29.00-30.00   sec  41.8 MBytes  351 Mb/s     31336
[ 5] 30.00-31.00   sec  43.5 MBytes  365 Mb/s     32569
[ 5] 31.00-32.00   sec  45.7 MBytes  383 Mb/s     34241
[ 5] 32.00-33.00   sec  46.5 MBytes  390 Mb/s     34854
[ 5] 33.00-34.00   sec  43.9 MBytes  368 Mb/s     32847
[ 5] 34.00-35.00   sec  44.4 MBytes  373 Mb/s     33288
[ 5] 35.00-36.00   sec  46.6 MBytes  391 Mb/s     34938
[ 5] 36.00-37.00   sec  45.4 MBytes  381 Mb/s     33984
[ 5] 37.00-38.00   sec  43.2 MBytes  362 Mb/s     32364
[ 5] 38.00-39.00   sec  45.8 MBytes  384 Mb/s     34320
[ 5] 39.00-40.00   sec  44.7 MBytes  374 Mb/s     33489
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00    sec  1.75 GBytes  376 Mb/s     0.000 ms    0/1343630 (0%) sender
[ 5]  0.00-40.00    sec  1.74 GBytes  375 Mb/s     0.048 ms   5513/1343630 (0.41%) receiver

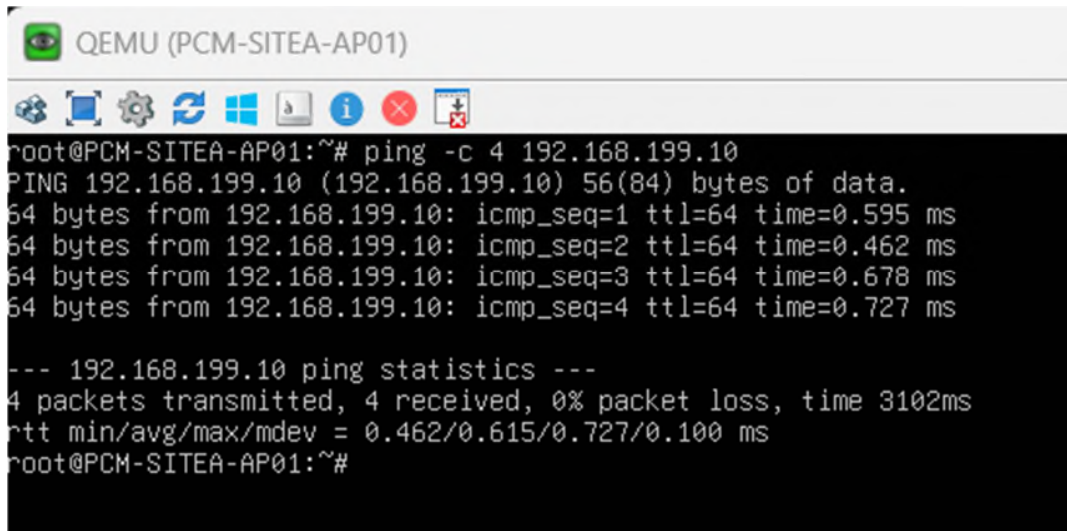
iperf Done.
root@PCM-SITED-SW01:~#

```

Figure 15 – screenshot of PCM-SITED-SW01 speed test

PCM-SITEA-AP01

PCM-SITEA-AP01 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEA-AP01)". The terminal displays the output of a ping command: `ping -c 4 192.168.199.10`. The output shows four successful ping requests with varying response times. Below the individual results, a summary line indicates "4 packets transmitted, 4 received, 0% packet loss, time 3102ms". The final line shows the round-trip time statistics: `rtt min/avg/max/mdev = 0.462/0.615/0.727/0.100 ms`.

```
root@PCM-SITEA-AP01:~# ping -c 4 192.168.199.10
PING 192.168.199.10 (192.168.199.10) 56(84) bytes of data.
64 bytes from 192.168.199.10: icmp_seq=1 ttl=64 time=0.595 ms
64 bytes from 192.168.199.10: icmp_seq=2 ttl=64 time=0.462 ms
64 bytes from 192.168.199.10: icmp_seq=3 ttl=64 time=0.678 ms
64 bytes from 192.168.199.10: icmp_seq=4 ttl=64 time=0.727 ms

--- 192.168.199.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3102ms
rtt min/avg/max/mdev = 0.462/0.615/0.727/0.100 ms
root@PCM-SITEA-AP01:~#
```

Figure 16 – screenshot of PCM-SITEA-AP01 ping test

PCM-SITEA-AP02 Speed Test

QEMU (PCM-SITEA-AP01)

[ID]	Interval		Transfer	Bitrate	Retr	Cwnd
[5]	0.00-1.00	sec	91.5 MBytes	767 Mbits/sec	0	7.15 MBytes
[5]	1.00-2.00	sec	104 MBytes	872 Mbits/sec	0	7.15 MBytes
[5]	2.00-3.00	sec	104 MBytes	875 Mbits/sec	0	7.15 MBytes
[5]	3.00-4.00	sec	102 MBytes	860 Mbits/sec	0	7.15 MBytes
[5]	4.00-5.00	sec	104 MBytes	874 Mbits/sec	0	7.15 MBytes
[5]	5.00-6.00	sec	102 MBytes	859 Mbits/sec	0	7.15 MBytes
[5]	6.00-7.00	sec	103 MBytes	864 Mbits/sec	0	7.15 MBytes
[5]	7.00-8.00	sec	100 MBytes	840 Mbits/sec	0	7.15 MBytes
[5]	8.00-9.00	sec	103 MBytes	866 Mbits/sec	0	7.15 MBytes
[5]	9.00-10.00	sec	103 MBytes	865 Mbits/sec	0	7.15 MBytes
[5]	10.00-11.00	sec	96.0 MBytes	806 Mbits/sec	0	7.15 MBytes
[5]	11.00-12.00	sec	104 MBytes	872 Mbits/sec	0	7.15 MBytes
[5]	12.00-13.00	sec	100 MBytes	842 Mbits/sec	0	7.15 MBytes
[5]	13.00-14.00	sec	104 MBytes	873 Mbits/sec	0	7.15 MBytes
[5]	14.00-15.00	sec	104 MBytes	873 Mbits/sec	0	7.15 MBytes
[5]	15.00-16.00	sec	104 MBytes	871 Mbits/sec	0	7.15 MBytes
[5]	16.00-17.00	sec	104 MBytes	875 Mbits/sec	0	7.15 MBytes
[5]	17.00-18.00	sec	102 MBytes	852 Mbits/sec	0	7.15 MBytes
[5]	18.00-19.00	sec	104 MBytes	873 Mbits/sec	0	7.15 MBytes
[5]	19.00-20.00	sec	104 MBytes	872 Mbits/sec	0	7.15 MBytes
[5]	20.00-21.00	sec	100 MBytes	838 Mbits/sec	0	7.15 MBytes
[5]	21.00-22.00	sec	104 MBytes	874 Mbits/sec	0	7.15 MBytes
[5]	22.00-23.00	sec	104 MBytes	871 Mbits/sec	0	7.15 MBytes
[5]	23.00-24.00	sec	103 MBytes	863 Mbits/sec	0	7.15 MBytes
[5]	24.00-25.00	sec	104 MBytes	871 Mbits/sec	0	7.15 MBytes
[5]	25.00-26.00	sec	100 MBytes	841 Mbits/sec	0	7.15 MBytes
[5]	26.00-27.00	sec	103 MBytes	861 Mbits/sec	0	7.15 MBytes
[5]	27.00-28.00	sec	101 MBytes	850 Mbits/sec	0	7.15 MBytes
[5]	28.00-29.00	sec	104 MBytes	872 Mbits/sec	0	7.15 MBytes
[5]	29.00-30.00	sec	104 MBytes	875 Mbits/sec	0	7.15 MBytes
[5]	30.00-31.00	sec	103 MBytes	862 Mbits/sec	0	7.15 MBytes
[5]	31.00-32.00	sec	102 MBytes	860 Mbits/sec	0	7.15 MBytes
[5]	32.00-33.00	sec	104 MBytes	868 Mbits/sec	0	7.15 MBytes
[5]	33.00-34.00	sec	104 MBytes	872 Mbits/sec	0	7.15 MBytes
[5]	34.00-35.00	sec	104 MBytes	874 Mbits/sec	0	7.15 MBytes
[5]	35.00-36.00	sec	104 MBytes	873 Mbits/sec	0	7.15 MBytes
[5]	36.00-37.00	sec	104 MBytes	875 Mbits/sec	0	7.15 MBytes
[5]	37.00-38.00	sec	103 MBytes	867 Mbits/sec	0	7.15 MBytes
[5]	38.00-39.00	sec	104 MBytes	877 Mbits/sec	0	7.15 MBytes
[5]	39.00-40.00	sec	103 MBytes	862 Mbits/sec	0	7.15 MBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[5]	0.00-40.00	sec	4.01 GBytes	861 Mbits/sec	0	sender
[5]	0.00-40.01	sec	4.01 GBytes	861 Mbits/sec		receiver

iperf Done.
root@PCM-SITEA-AP01:~#

Figure 17 – screenshot of PCM-SITEA-AP01 speed test

PCM-SITEA-AP01 Jitter Test

```

QEMU (PCM-SITEA-AP01)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5]  0.00-1.00    sec  46.3 MBytes  388 Mbits/sec  34710
[ 5]  1.00-2.00    sec  46.8 MBytes  392 Mbits/sec  35015
[ 5]  2.00-3.00    sec  46.0 MBytes  386 Mbits/sec  34458
[ 5]  3.00-4.00    sec  46.6 MBytes  391 Mbits/sec  34937
[ 5]  4.00-5.00    sec  46.4 MBytes  389 Mbits/sec  34766
[ 5]  5.00-6.00    sec  43.6 MBytes  366 Mbits/sec  32678
[ 5]  6.00-7.00    sec  47.5 MBytes  399 Mbits/sec  35543
[ 5]  7.00-8.00    sec  47.3 MBytes  397 Mbits/sec  35435
[ 5]  8.00-9.00    sec  45.6 MBytes  382 Mbits/sec  34136
[ 5]  9.00-10.00   sec  47.5 MBytes  399 Mbits/sec  35599
[ 5] 10.00-11.00   sec  47.1 MBytes  396 Mbits/sec  35314
[ 5] 11.00-12.00   sec  42.6 MBytes  358 Mbits/sec  31924
[ 5] 12.00-13.00   sec  44.9 MBytes  376 Mbits/sec  33625
[ 5] 13.00-14.00   sec  44.1 MBytes  371 Mbits/sec  33052
[ 5] 14.00-15.00   sec  45.8 MBytes  384 Mbits/sec  34330
[ 5] 15.00-16.00   sec  46.6 MBytes  391 Mbits/sec  34911
[ 5] 16.00-17.00   sec  44.3 MBytes  372 Mbits/sec  33201
[ 5] 17.00-18.00   sec  44.6 MBytes  374 Mbits/sec  33390
[ 5] 18.00-19.00   sec  46.2 MBytes  388 Mbits/sec  34569
[ 5] 19.00-20.00   sec  47.8 MBytes  401 Mbits/sec  35783
[ 5] 20.00-21.00   sec  45.9 MBytes  385 Mbits/sec  34352
[ 5] 21.00-22.01   sec  44.7 MBytes  371 Mbits/sec  33493
[ 5] 22.01-23.00   sec  46.3 MBytes  393 Mbits/sec  34669
[ 5] 23.00-24.00   sec  41.7 MBytes  349 Mbits/sec  31201
[ 5] 24.00-25.00   sec  43.9 MBytes  369 Mbits/sec  32917
[ 5] 25.00-26.00   sec  44.6 MBytes  374 Mbits/sec  33392
[ 5] 26.00-27.00   sec  44.1 MBytes  370 Mbits/sec  32993
[ 5] 27.00-28.00   sec  45.3 MBytes  380 Mbits/sec  33921
[ 5] 28.00-29.00   sec  42.4 MBytes  355 Mbits/sec  31721
[ 5] 29.00-30.00   sec  42.9 MBytes  360 Mbits/sec  32137
[ 5] 30.00-31.00   sec  44.2 MBytes  371 Mbits/sec  33140
[ 5] 31.00-32.00   sec  40.9 MBytes  343 Mbits/sec  30657
[ 5] 32.00-33.00   sec  44.6 MBytes  374 Mbits/sec  33377
[ 5] 33.00-34.00   sec  45.9 MBytes  385 Mbits/sec  34376
[ 5] 34.00-35.00   sec  42.4 MBytes  356 Mbits/sec  31770
[ 5] 35.00-36.00   sec  45.2 MBytes  379 Mbits/sec  33867
[ 5] 36.00-37.00   sec  41.6 MBytes  349 Mbits/sec  31150
[ 5] 37.00-38.00   sec  46.2 MBytes  388 Mbits/sec  34610
[ 5] 38.00-39.00   sec  47.2 MBytes  396 Mbits/sec  35380
[ 5] 39.00-40.00   sec  44.6 MBytes  374 Mbits/sec  33439
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5]  0.00-40.00   sec  1.76 GBytes  378 Mbits/sec  0.000 ms    0/1349938 (0%) sender
[ 5]  0.00-40.00   sec  1.75 GBytes  377 Mbits/sec  0.045 ms   5051/1349938 (0.37%) receiver

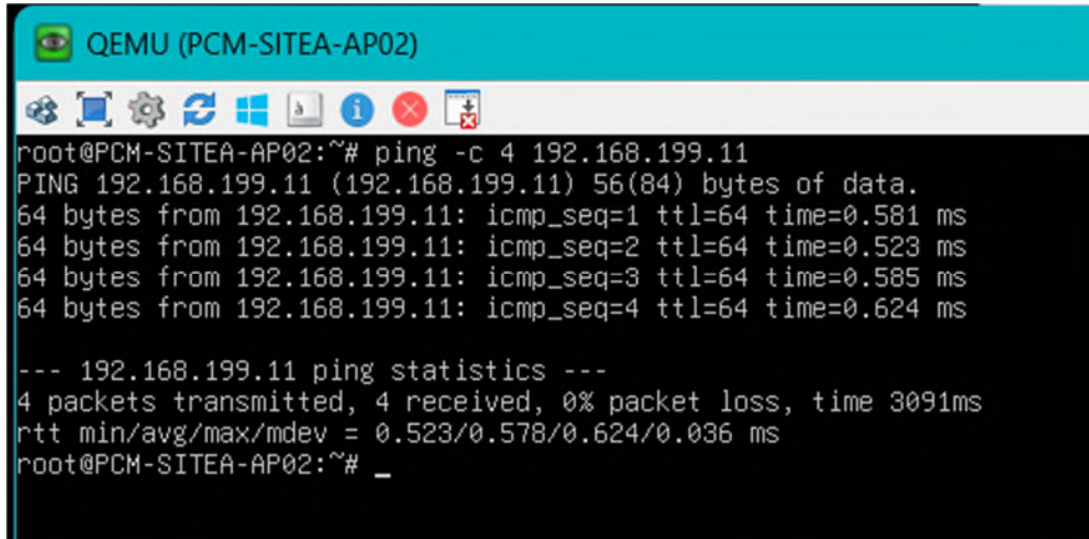
iperf Done.
root@PCM-SITEA-AP01:~#

```

Figure 18 – screenshot of PCM-SITEA-AP01 jitter test

PCM-SITEA-AP02

PCM-SITEA-AP02 Ping Test



The screenshot shows a QEMU terminal window titled "QEMU (PCM-SITEA-AP02)". The terminal displays the output of a ping command executed from the root user at the PCM-SITEA-AP02 host. The command is "ping -c 4 192.168.199.11". The output shows four successful ping requests, each with a TTL of 64 and a response time between 0.523 ms and 0.624 ms. The statistics section indicates that 4 packets were transmitted and received with 0% packet loss and a total time of 3091ms. The round-trip time (rtt) statistics are: min/avg/max/mdev = 0.523/0.578/0.624/0.036 ms.

```
root@PCM-SITEA-AP02:~# ping -c 4 192.168.199.11
PING 192.168.199.11 (192.168.199.11) 56(84) bytes of data.
64 bytes from 192.168.199.11: icmp_seq=1 ttl=64 time=0.581 ms
64 bytes from 192.168.199.11: icmp_seq=2 ttl=64 time=0.523 ms
64 bytes from 192.168.199.11: icmp_seq=3 ttl=64 time=0.585 ms
64 bytes from 192.168.199.11: icmp_seq=4 ttl=64 time=0.624 ms

--- 192.168.199.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3091ms
rtt min/avg/max/mdev = 0.523/0.578/0.624/0.036 ms
root@PCM-SITEA-AP02:~# _
```

Figure 19 – screenshot of PCM-SITEA-AP02 ping test

PCM-SITEA-AP02 Speed Test

```

QEMU (PCM-SITEA-AP02)

[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[  5]  0.00-1.00    sec    103 MBytes    861 Mbits/sec    0   426 KBytes
[  5]  1.00-2.00    sec    105 MBytes    881 Mbits/sec    0   675 KBytes
[  5]  2.00-3.00    sec    104 MBytes    868 Mbits/sec    0   744 KBytes
[  5]  3.00-4.00    sec    103 MBytes    861 Mbits/sec    0   744 KBytes
[  5]  4.00-5.00    sec    104 MBytes    871 Mbits/sec    0   744 KBytes
[  5]  5.00-6.00    sec    98.8 MBytes    829 Mbits/sec    0   744 KBytes
[  5]  6.00-7.00    sec    104 MBytes    871 Mbits/sec    0   744 KBytes
[  5]  7.00-8.00    sec    103 MBytes    862 Mbits/sec    0   744 KBytes
[  5]  8.00-9.00    sec    103 MBytes    867 Mbits/sec    0   782 KBytes
[  5]  9.00-10.00   sec    103 MBytes    862 Mbits/sec    0   782 KBytes
[  5] 10.00-11.00   sec    104 MBytes    873 Mbits/sec    0   782 KBytes
[  5] 11.00-12.00   sec    103 MBytes    862 Mbits/sec    0   782 KBytes
[  5] 12.00-13.00   sec    104 MBytes    873 Mbits/sec    0   782 KBytes
[  5] 13.00-14.00   sec    103 MBytes    863 Mbits/sec    0   782 KBytes
[  5] 14.00-15.00   sec    104 MBytes    875 Mbits/sec    0   782 KBytes
[  5] 15.00-16.00   sec    88.0 MBytes    738 Mbits/sec    0   1.14 MBytes
[  5] 16.00-17.00   sec    103 MBytes    863 Mbits/sec    0   1.14 MBytes
[  5] 17.00-18.00   sec    103 MBytes    862 Mbits/sec    0   1.14 MBytes
[  5] 18.00-19.00   sec    103 MBytes    863 Mbits/sec    0   1.14 MBytes
[  5] 19.00-20.00   sec    99.0 MBytes    830 Mbits/sec    0   1.14 MBytes
[  5] 20.00-21.00   sec    99.1 MBytes    832 Mbits/sec    0   1.14 MBytes
[  5] 21.00-22.00   sec    103 MBytes    863 Mbits/sec    0   1.14 MBytes
[  5] 22.00-23.00   sec    103 MBytes    867 Mbits/sec    0   1.14 MBytes
[  5] 23.00-24.00   sec    102 MBytes    852 Mbits/sec    0   1.14 MBytes
[  5] 24.00-25.00   sec    103 MBytes    862 Mbits/sec    0   1.14 MBytes
[  5] 25.00-26.00   sec    99.0 MBytes    830 Mbits/sec    0   1.14 MBytes
[  5] 26.00-27.00   sec    103 MBytes    863 Mbits/sec    0   1.14 MBytes
[  5] 27.00-28.00   sec    103 MBytes    865 Mbits/sec    0   1.14 MBytes
[  5] 28.00-29.00   sec    103 MBytes    864 Mbits/sec    0   1.14 MBytes
[  5] 29.00-30.00   sec    104 MBytes    876 Mbits/sec    0   1.14 MBytes
[  5] 30.00-31.00   sec    97.9 MBytes    821 Mbits/sec    0   1.14 MBytes
[  5] 31.00-32.00   sec    104 MBytes    874 Mbits/sec    0   1.14 MBytes
[  5] 32.00-33.00   sec    103 MBytes    862 Mbits/sec    0   1.14 MBytes
[  5] 33.00-34.00   sec    103 MBytes    863 Mbits/sec    0   1.71 MBytes
[  5] 34.00-35.00   sec    103 MBytes    864 Mbits/sec    0   1.71 MBytes
[  5] 35.00-36.00   sec    103 MBytes    862 Mbits/sec    0   1.71 MBytes
[  5] 36.00-37.00   sec    103 MBytes    862 Mbits/sec    0   1.71 MBytes
[  5] 37.00-38.00   sec    98.1 MBytes    823 Mbits/sec    0   1.71 MBytes
[  5] 38.00-39.00   sec    102 MBytes    852 Mbits/sec    0   1.71 MBytes
[  5] 39.00-40.00   sec    102 MBytes    853 Mbits/sec    0   1.71 MBytes

[ ID] Interval      Transfer      Bitrate      Retr
[  5]  0.00-40.00   sec    3.99 GBytes    856 Mbits/sec    0
[  5]  0.00-40.01   sec    3.98 GBytes    856 Mbits/sec    0

iperf Done.
root@PCM-SITEA-AP02:~#

```

Figure 20 – screenshot of PCM-SITEA-AP02 speed test

PCM-SITEA-AP02 Jitter Test

```

QEMU (PCM-SITEA-AP02)
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec 44.4 MBytes 372 Mbits/sec 33240
[ 5] 1.00-2.00 sec 42.9 MBytes 360 Mbits/sec 32143
[ 5] 2.00-3.00 sec 45.0 MBytes 377 Mbits/sec 33694
[ 5] 3.00-4.00 sec 46.0 MBytes 386 Mbits/sec 34458
[ 5] 4.00-5.01 sec 44.5 MBytes 370 Mbits/sec 33319
[ 5] 5.01-6.00 sec 47.0 MBytes 397 Mbits/sec 35168
[ 5] 6.00-7.00 sec 47.5 MBytes 399 Mbits/sec 35593
[ 5] 7.00-8.00 sec 46.7 MBytes 391 Mbits/sec 34955
[ 5] 8.00-9.00 sec 47.2 MBytes 396 Mbits/sec 35367
[ 5] 9.00-10.00 sec 45.1 MBytes 379 Mbits/sec 33791
[ 5] 10.00-11.00 sec 43.7 MBytes 367 Mbits/sec 32733
[ 5] 11.00-12.00 sec 41.7 MBytes 350 Mbits/sec 31256
[ 5] 12.00-13.00 sec 43.5 MBytes 365 Mbits/sec 32599
[ 5] 13.00-14.00 sec 42.9 MBytes 360 Mbits/sec 32160
[ 5] 14.00-15.00 sec 45.8 MBytes 384 Mbits/sec 34276
[ 5] 15.00-16.00 sec 43.2 MBytes 363 Mbits/sec 32390
[ 5] 16.00-17.00 sec 43.7 MBytes 367 Mbits/sec 32765
[ 5] 17.00-18.00 sec 47.9 MBytes 401 Mbits/sec 35860
[ 5] 18.00-19.00 sec 47.6 MBytes 400 Mbits/sec 35657
[ 5] 19.00-20.00 sec 47.8 MBytes 401 Mbits/sec 35804
[ 5] 20.00-21.00 sec 45.2 MBytes 379 Mbits/sec 33841
[ 5] 21.00-22.00 sec 45.9 MBytes 385 Mbits/sec 34361
[ 5] 22.00-23.00 sec 43.5 MBytes 365 Mbits/sec 32575
[ 5] 23.00-24.00 sec 42.1 MBytes 353 Mbits/sec 31537
[ 5] 24.00-25.00 sec 45.6 MBytes 382 Mbits/sec 34124
[ 5] 25.00-26.00 sec 43.8 MBytes 367 Mbits/sec 32783
[ 5] 26.00-27.00 sec 44.0 MBytes 369 Mbits/sec 32963
[ 5] 27.00-28.00 sec 45.3 MBytes 380 Mbits/sec 33930
[ 5] 28.00-29.00 sec 45.0 MBytes 378 Mbits/sec 33741
[ 5] 29.00-30.00 sec 43.4 MBytes 364 Mbits/sec 32518
[ 5] 30.00-31.00 sec 44.8 MBytes 376 Mbits/sec 33540
[ 5] 31.00-32.00 sec 44.5 MBytes 374 Mbits/sec 33360
[ 5] 32.00-33.00 sec 44.4 MBytes 373 Mbits/sec 33279
[ 5] 33.00-34.00 sec 45.5 MBytes 382 Mbits/sec 34112
[ 5] 34.00-35.00 sec 45.9 MBytes 386 Mbits/sec 34398
[ 5] 35.00-36.00 sec 45.4 MBytes 381 Mbits/sec 34026
[ 5] 36.00-37.00 sec 40.6 MBytes 341 Mbits/sec 30415
[ 5] 37.00-38.00 sec 43.5 MBytes 365 Mbits/sec 32600
[ 5] 38.00-39.00 sec 45.3 MBytes 380 Mbits/sec 33942
[ 5] 39.00-40.00 sec 44.3 MBytes 372 Mbits/sec 33197

-- -- -- -- --
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-40.00 sec 1.75 GBytes 376 Mbits/sec 0.000 ms 0/1342470 (0%) sender
[ 5] 0.00-40.00 sec 1.75 GBytes 375 Mbits/sec 0.045 ms 3681/1342470 (0.27%) receiver

iperf Done.
root@PCM-SITEA-AP02:~#

```

Figure 21 – screenshot of PCM-SITEA-AP02 jitter test